

Group-aware validation removes most of the apparent accuracy of extraction-category classification in public cannabis certificates of analysis

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Preprint. Code and evidence logs: <https://github.com/ArtBreguez/cannabis-extraction-ds>

Abstract

Published models that classify cannabis products from cannabinoid and terpene profiles report near-perfect accuracy: above 95% for resin type, 100% for chemovar class. These figures come from random cross-validation or Kennard-Stone partitioning, which hold out no plant, producer or laboratory, so replicates may span the train/test boundary.

We quantify that cost on a task whose group structure is explicit. On 33,229 certificate rows of cannabis concentrates that name a producer, each with a source-declared extraction category (non-solvent against solvent-based), a gradient-boosted classifier scores 78.5% balanced accuracy (ROC AUC 0.89) under random five-fold cross-validation. Holding out the producer leaves 54.0% (AUC 0.62). The 24-point loss shrinks to 11 for a logistic regression, and class weighting lifts the held-out score to 56 to 61%. A producer-level bootstrap puts the held-out AUC between 0.54 and 0.70: a modest signal survives, less than a third of what the random split shows above chance.

A designed experiment shows the same direction: 162 HPLC-assayed samples fall from 56.7% under random folds to 38.9% under leave-one-variety-out with gradient boosting, and from 54.3% to 46.3% with logistic regression, against 33.3% chance. In the market data the same profiles identify the producer at 67.1% against a 14.4% baseline and the laboratory at 92.3% against 40.3%.

We conclude that the declared extraction category is only weakly recoverable from the analyte panels of this corpus for an unseen producer, and that a random split overstates it severalfold. We did not rerun the published studies and make no claim about their chemical conclusions, only that their figures come from validation that holds out no group.

Keywords: cannabis, chemometrics, data leakage, group cross-validation, extraction method, certificates of analysis, negative result

1. Introduction

Cannabis chemical profiles are now routinely fed to supervised classifiers. Reported performance is consistently high. A support vector machine separates Dutch-type from Moroccan-type cannabis resin at 95.2% accuracy from two cannabinoid concentrations [1]. Partial least squares discriminant analysis as-

signs chemovar class from near-infrared spectra of whole inflorescences with "100% prediction accuracy", sensitivity and specificity both 1.00 [2]. A combined cannabinoid-and-terpene PLS-DA model reaches 0% prediction misclassification across chemovars [3].

These numbers invite a question the papers do not ask: performance on *what* unseen unit? A shuffled fold defines "unseen" as a row absent from the training partition. Deployment defines it more strictly, as a plant, a producer, or a laboratory the model has never encountered. When several rows descend from the same entity, a random split lets the model recognise the entity instead of learning the property of interest, and the reported score measures memorisation.

This failure mode is established in the general methodological literature [4, 5, 6, 7]; a related form, selection bias from choosing features outside the cross-validation loop, was quantified in genomics more than two decades ago [11]. It has been described specifically for chemometrics, where Király and Tóth show that the Kennard-Stone selection algorithm and the inappropriate handling of repeated measurements both introduce train/test leakage [8]. The cannabis applications cited above are exposed to it in different ways. Reference [2] grows 88 genotypes cloned in triplicate to obtain 264 plants, then partitions with Kennard-Stone [13] with no statement that the three clones of a genotype were kept in one partition, on a 231/33 class split. Reference [1] evaluates plain accuracy over five random 80/20 train/test iterations on a 149/40 class split, where always answering the majority class already scores 78.8%; our criticism there is of the metric, since the paper does not say whether its samples share sources. Reference [3] reports a 4% cross-validation error beside its 0% prediction error and does not describe how its prediction set was drawn.

The bridge between the two literatures has not been built. No published cannabis study that we could locate holds out the producer, the genotype or the laboratory and reports what happens to the score.

We build that bridge on a concrete task. The question is whether a concentrate's cannabinoid and terpene profile carries enough information to recover how it was made, separating concentrates declared *non-solvent based*, which under the Nevada rule in force for the period means mechanical separation (hash, kief, rosin) or extraction with ethanol or CO₂ [12], from those declared *solvent based*, made with any other approved solvent, in practice hydrocarbons. We keep the shorthand `solventless` and `hydrocarbon` for the two declared categories because the code and evidence logs use them; 2.1 reports what each category contains. The task is a good test case for three reasons. The label is declared at source rather than inferred. The group structure is explicit, since every sample names a producer and a laboratory. And a substantial subset of producers make both classes, which permits a test where a held-out producer's label cannot be read off the training rows.

We report a largely negative result for the chemical question and a quantitative result for the methodological one.

2. Materials and methods

2.1 Data

Market data. The Cannlytics `cannabis_results` corpus [9] aggregates public cannabis certificates of analysis obtained from state regulatory portals and published COAs, released under CC-BY-4.0. The retrieved file contains 808,407 rows, of which 139,714 are concentrates or extracts.

Labels come from the `product_type` field, which in some markets declares the extraction family directly as `non-solvent based concentrate` or `solvent based concentrate`. This is a source-declared label, recorded on the certificate, not an inference from a commercial product name. Rows whose `product_type` is only `concentrate` remain unlabelled and are never assigned to a class. The analyses run on the 37,344 labelled rows that survive the consistency check described below, which we call the usable rows.

Every usable row that carries a `lab_state` is from Nevada: seven laboratories and 31,422 of the usable rows (84.1%). The field is blank for the remaining 5,922 rows, which come from five laboratories, four that record no producer and one that does. No row names another state, but the names of the four producer-less laboratories (one is `Cannalytics RI`) suggest they are not in Nevada, in which case the rule cited below does not govern their 4,115 rows. Tests are dated 2020-01-02 to 2024-03-29 on the 33,765 rows that carry a date.

Rows are not samples. Of the usable rows, 581 are exact duplicates and 1,813 (4.9%) repeat another row's product name, laboratory, producer and all 19 analyte values. Separately, 1,303 samples from the producer-less laboratories are stored as two adjacent rows each, with the same laboratory and date and complementary analytes (`delta_9_thc` in one; `cbd`, `cbda`, `thca` and `total_cbd` in the other), so those laboratories' 4,115 rows describe 2,812 samples. We keep the rows as stored, since nothing marks which copy is the record, and 3.2 reports what dropping the duplicates and merging the pairs does. The split pairs lie outside the 33,229 rows that name a producer, on which the headline comparison runs. Those rows come from 96 producers of very unequal size: the largest holds 13.2% of them, the five largest 42.3%, and the median producer 64 rows.

The two categories are regulatory, not chemical. Nevada's rule separates the same pair, and in its versions of July 2022 and October 2023 defines it as "Extract of cannabis (nonsolvent) like hashish, bubble hash, infused dairy butter, mixtures of extracted products or oils or fats derived from natural sources, including concentrated cannabis extracted with ethanol or CO2" against "Extract of cannabis (solvent-based) made with any approved solvent, including concentrated cannabis extracted by means other than with ethanol or CO2" [12]. The text effective November 2020 already carries the second definition, and the 2018 rule it replaced named CO2 only. Under the current definition the non-solvent category spans mechanical separation, ethanol and CO2, and the product names are consistent with that. Table 1 counts them over the rows that carry a name, which are the 33,229 rows with a producer; a name may hit several families.

Table 1. Product-name word families in each declared category, over the 33,229 named rows.

word family in product_name	non-solvent	solvent-based
distillate	5,805 (51.5%)	8,258 (37.6%)
co2 / supercritical	225 (2.0%)	19 (0.1%)
ethanol	163 (1.4%)	214 (1.0%)
hydrocarbon words (bho, butane, live resin, shatter, badder, wax)	43 (0.4%)	3,805 (17.3%)
mechanical words (rosin, hash, kief, bubble, ice water)	559 (5.0%)	114 (0.5%)

Only 5.0% of the named non-solvent rows contain a mechanical word, while 51.5% are named `distillate`, a product that presupposes a crude extract and is not how mechanical concentrates are sold, 2.0% name CO2 and 1.4% name ethanol. The word families are crude. All 43 non-solvent rows that hit a hydro-

carbon word also say rosin, where badder and wax are consistencies, and the 114 solvent-based rows that hit a mechanical word contain hash, bubble or kief inside cultivar names, none of them rosin. The table describes the two categories; it cannot count mislabelled rows.

Neither the wording of the rule nor practice was constant over the period, and they did not change together. Product names that say ethanol are solvent-based through 2022 and non-solvent in 2023 (Table 2), so the meaning of the label drifts within the data.

Table 2. Rows whose product name says ethanol, by year of test and declared category.

year	solvent-based	non-solvent
2020	82	9
2021	21	1
2022	98	0
2023	13	133

The task this paper tests is therefore the regulator's category as the laboratories applied it over four years, mechanical, ethanol or CO2 extraction against the remaining solvents, and not rosin against BHO specifically; 3.2 reports the headline pair on two narrower subsets. The hydrocarbon shorthand is close to the current rule's meaning, since the approved solvents other than ethanol and CO2 are hydrocarbons.

Most concentrate rows carry no such declaration and are therefore dropped at this step, which is where the bulk of the reduction happens:

Table 3. From concentrate rows to labelled rows.

concentrate or extract rows	139,714	
product_type declares the extraction family	37,310	(26.7%)
product_type does not	102,404	(73.3%)
labelled rows outside that subset	64	
total labelled rows	37,374	

The discarded majority is dominated by values that name a format rather than a method: extracts (34,034 rows), concentrate, product inhalable (30,586), bare concentrate (26,030) and marijuana extract for inhalation (3,901). These are not missing labels to be recovered, since the extraction family was never recorded, so imputing one would invent the dependent variable. Matching is by regular expression (`non[- ]?solvent`) rather than exact string, which also admits regional variants such as non-solvent concentrate (1,445 rows) and, in 64 cases, infused pre-rolls (non-solvent). Those 64 rows are the reason the labelled total slightly exceeds the declared count within the concentrate subset; they are solventless by declaration but are pre-rolls rather than concentrates, and they are 0.17% of the labelled data.

A secondary labelling path applies regular expressions to product_name (hash rosin, live rosin, rosin for solventless; live resin, bho, shatter, badder, wax for hydrocarbon). The two paths can only be compared on the 11.5% of labelled rows whose names carry one of those words. Where both paths produce a label they agree on 4,254 of 4,284 rows, 99.3%. The 30 contradictory rows are excluded. The name path never adds a row: it only cross-checks the declared label, so every labelled row carries a product_type declaration.

The resulting labelled dataset is 37,374 rows, of which 30 carry contradictory labels and are excluded from every analysis, leaving **37,344 rows** that all figures below are computed on unless stated otherwise:

Table 4. Class counts on the usable rows.

hydrocarbon	24,573	65.8%
solventless	12,771	34.2%

Grouping variables are populated as follows:

Table 5. Grouping variables on the usable rows.

producer	33,229 rows (89.0%)	96 distinct
lab	37,344 rows (100.0%)	12 distinct
strain_name	0 rows (0.0%)	0 distinct

Controlled data. Zenodo record 13823859 [10] is a designed experiment rather than market data: six named varieties of non-psychoactive, CBD-type cannabis, each extracted by three laboratory methods (maceration, ultrasound, supercritical CO₂) under varied time, temperature and pressure, with cannabinoids quantified by HPLC. The design is fully balanced at 54 samples per method and 27 per variety, 162 total, with four features (CBD, CBG, CBN, THC) and no missing values. Each sample is a distinct (method, variety, time, temperature, pressure) cell, so there are no replicates to leak across a random split; the chemotype also limits transfer to THC-dominant concentrates. The process settings (time, temperature, pressure) are not features: pressure alone identifies supercritical CO₂, and adding the three settings lifts the random split to 88.3%.

2.2 Feature construction

The source file carries 34 analyte columns. Fifteen are empty for every labelled row and are dropped rather than imputed, since imputing a never-populated column fabricates a feature. Nineteen analytes remain.

Non-detects require care. The corpus has already normalised **ND** and **<LOQ** strings away, so no such strings survive, which collapses a four-way distinction into a two-way one:

- a numeric **zero** is a real measurement at or below the detection limit;
- an **empty** cell is an absence of information, either not tested or not reported.

These are not the same and must not be merged. The labelled dataset therefore keeps two columns per analyte, the value and a boolean **_tested** mask, and rows are never dropped for having gaps. The classifier receives the 19 value columns with empty cells left missing, which the estimator routes natively at each split, so the tested-versus-unttested distinction reaches the model through missingness rather than through the mask columns. The mask columns drive the coverage figures below and the pre-check in 2.3.

This distinction produces two different and both meaningful coverage figures, which we separate explicitly because conflating them is easy:

Table 6. Share of rows with the analyte tested and with a non-zero value, by class.

analyte	tested (flag)		detected (value > 0)	
	solventless	hydro	solventless	hydro
d_limonene	75.7%	81.1%	40.7%	65.6%
beta_caryophyllene	75.7%	81.1%	47.0%	68.3%
alpha_pinene	75.7%	81.1%	34.2%	56.3%
total_terpenes	88.2%	89.4%	56.8%	74.5%

Terpene panels are requested at similar rates for both classes, a gap of 1 to 6 points. The large gap, 18 to 25 points, is in *detection*: hydrocarbon extracts report a non-zero terpene value far more often. That gap is in what is reported once a panel is run, not in which test was ordered, and with 51.5% of the named non-solvent rows being distillates, refining is a likelier cause than the extraction category itself. The panel is not constant over time either. Nevada listed terpene analysis among the required tests for both categories in the text effective November 2020 and no longer does in the July 2022 text [12], and the share of rows with a terpene panel falls from 96.2% in 2020 to 84.9% in 2022, 47.7% in 2023 and none in 2024.

Two properties of these columns are worth stating because they bound what the features can mean.

First, `total_terpenes` is not an independent measurement. It is a near-exact sum of the ten terpenes that survive the empty-column drop ($R^2 = 0.984$ against their sum, Pearson $r = 0.992$), so it is collinear with features already in the matrix and any per-feature importance attributed to it is split credit rather than chemistry. Its residual against that sum is laboratory-specific: `CERTIFIED AG LAB` reports an exact sum (within 10^{-6}) on 99.7% of rows, `MA & ASSOCIATES` on none, which makes the residual a laboratory fingerprint in its own right. `total_thc` and `total_cbd` are derived columns of the same kind, each a fixed combination of its acid and neutral forms. Dropping the column costs 0.3 points of balanced accuracy, so nothing in this paper depends on keeping it; we keep it because the published literature does.

Second, whether `total_terpenes` is populated at all is a laboratory convention, not a per-sample decision. Every laboratory is at 0% or 100%, with no intermediate value, and the populated set is *exactly* the set of rows that carry a producer:

```
total terpenes populated == producer present: 33,229 / 33,229 rows, no exceptions
laboratories at 100%: G3, NV CANN, DB, CERTIFIED AG, DPL NV, ERP, 374, MA
laboratories at 0%: PureVita, Cannalytics RI, Lifted Testing, Green Peaks
```

So `total_terpenes` being blank is an exact in-model indicator of "this row will be dropped by the unseen-producer scheme". That coupling is why 3.2 reports a same-sample comparison rather than resting on the headline gap.

Finally, 101 rows carry a `total_thc` above 100%, which is impossible as a percentage; the largest is 686,400, a milligram figure in a percent column. Ninety-four come from a single laboratory. They are 0.27% of the corpus and we leave them in place rather than silently editing source measurements, but they are unit artefacts rather than chemistry and no claim here rests on them.

2.3 The leakage pre-check

A detection gap of 18 to 25 points is a candidate shortcut. If the pattern of which terpenes were reported, or reported as non-zero, tracked the class, a classifier could score well while reading the reporting convention rather than the chemistry. We measured this before modelling rather than assuming it away.

A classifier given **only the missingness pattern**, with every measured value discarded, reaches 69.9% accuracy against a 65.8% majority-class baseline: a lift of **+4.1 points**. Adding laboratory identity as an explicit feature reaches 70.5%, a further 0.6 points. Those are in-sample figures for the lookup. Cross-validated, the probe scores 69.8% plain and 56.8% balanced accuracy, against 84.1% plain and 81.0% balanced for the main model under a random split, so the shortcut is 4.0 of 18.3 points on one metric and 6.8 of 31.0 on the other.

The probe sees exactly the 19 analyte columns the model sees; a probe narrower than the feature set would be a lower bound on the wrong quantity.

That 4.1-point lift is not spread evenly. The 19 columns take only 13 distinct missingness patterns, and 7 of them are **100% one class**:

Table 7. Missingness patterns that contain a single class, rows as stored.

```
patterns that are 100% one class: 7 of 13
rows they cover:                4123 (11.04%)
n= 1495  all solventless (6 of 19 analytes reported)
n= 1303  all hydrocarbon (1 of 19 analytes reported)
n= 1303  all hydrocarbon (4 of 19 analytes reported)
```

For 11% of the corpus the label is recoverable with certainty from which cells are blank, before any number is read. Almost all of these rows, 4,115 of 4,123, come from four laboratories (PureVita, Cannalytics RI, Lifted Testing, Green Peaks) that report a restricted analyte panel and record no producer, so the shortcut is a laboratory reporting convention and not chemistry. Much of it is an artefact of the split records of 2.1: the rows reporting one analyte and the rows reporting four are the two halves of the same 1,303 samples (1,230 of them at PureVita), all solvent-based, while the rows reporting six are all non-solvent. With each pair merged the columns take 12 patterns, 6 of them pure, covering 2,820 rows (7.82%), and the in-sample lookup scores 68.8% against a 64.6% majority. The remaining 8 rows sit in three sparsely-populated patterns from other laboratories and do carry a producer; 14 of the pure rows report no analyte at all and stay in the data. Those 4,115 rows are therefore also the rows the unseen-producer scheme must drop, which is why the same-sample comparison in 3.2 matters.

Laboratory and class are not degenerate, though they are far from independent. Only 373 rows, 1.0%, sit in laboratories that are more than 90% one class, and the two largest laboratories, 15,018 and 12,531 rows, both sit close to the overall 65.8/34.2 split; across all twelve the hydrocarbon share runs from 0.0% to 97.8%, and 3.2 gives the range for the eight largest.

The cheapest shortcut therefore buys about 4 points, concentrated in 11% of the rows and attributable to reporting convention. That is small enough that a model scoring in the eighties cannot be running on it alone, which is what made the main experiment worth running, but it is not nothing, and 3.2 reports what happens when those rows are excluded.

2.4 Model and split schemes

A single model produces every headline figure: HistGradientBoostingClassifier (scikit-learn), max_iter=200, seed 20260921, early stopping disabled, fed the 19 analyte columns of 2.2. The point of the study is the evaluation protocol, not the estimator, so the estimator is held constant across all schemes.

The alarm probes of 2.6 use the same estimator at `max_iter=120` and a learning rate of 0.05, for the reason given there. A standardised logistic regression (L2 penalty, $C = 1$) is run beside it in 3.2 and 3.3 as a low-capacity comparator. Since a linear model cannot route missing values, on the market data it receives $\log(1 + x)$ of each value with empty cells set to zero, plus one reported-or-not indicator per analyte. The headline model is unweighted, so 3.2 also runs both estimators with class weights inversely proportional to class frequency.

Four split schemes are compared on the market data:

1. **random**: stratified five-fold, a protocol of the kind used in the cannabis literature;
2. **unseen-producer**: `GroupKFold` grouped by producer, so no producer appears in both train and test;
3. **unseen-lab**: leave-one-lab-out. Each fold holds out one entire laboratory. Only laboratories with at least 500 rows are eligible to be held out, because a fold testing on a 100-row laboratory estimates nothing; 8 of the 12 laboratories clear that floor and all 8 contain both classes, giving 8 folds. The 4 excluded laboratories hold 561 rows, 1.50% of the data, and they are still present in every training set;
4. **dual-class + unseen**: restricted to the 53 producers that make *both* classes, grouped by producer.

Scheme 4 is the strictest available test. Restricting to producers who make both classes removes the degenerate case where a producer's identity determines its label outright, but it does not make producer identity uninformative: the solventless share across these 53 producers ranges from below 0.01 to 0.99 with a median of 0.35, and 30 of the 53 sit outside the 0.2 to 0.8 band. A classifier given *only* an encoded producer ID, and no chemistry at all, still reaches 84.4% balanced accuracy inside this subset under a random split. What the restriction guarantees is that the held-out producer's label cannot be inferred from having seen that same producer in training, since every producer in the subset contributes both classes. It covers 27,751 rows, 74.3% of the usable data, with a 17,312 / 10,439 class split.

On the controlled data, random five-fold is compared against leave-one-variety-out.

2.5 Metrics

The classes are roughly 1:2, so the extraction task is scored by balanced accuracy; where plain accuracy appears, in the 2.3 probe and the alarm probes of 2.6, it is given beside its majority-class baseline. Balanced accuracy has a chance level of 50.0% for two classes and 33.3% for three. Because it is computed at the default decision threshold, which an unweighted model on unbalanced classes does not place well, 3.2 also reports ROC AUC, which needs no threshold and has a chance level of 0.5. We also report macro F1, per-class recall, confusion matrices, and the spread across folds. Every "+/-" in a table is the population standard deviation of the fold scores (divisor n), while the t-intervals of 3.2 use the sample standard deviation, so the two do not convert directly. Two baselines are computed explicitly: most-frequent (balanced accuracy 50.0%) and stratified random (50.7% on the one draw made, 50.0% in expectation). The alarm probes of 2.6 are multi-class and report plain accuracy against the explicit majority-class baseline of their target, with balanced accuracy beside it.

Per-class recall matters here because a single aggregate figure hides which side of the problem the model actually solves.

2.6 Alarm models

Before the main task, two probes asked whether the chemistry predicts metadata it should not: the testing laboratory and the producer. A profile that identifies the laboratory is a profile that partially encodes instrumentation, calibration and reporting convention, which would compromise any score obtained while train and test share laboratories. A third target, cultivar, has no reliable field in this corpus and could only be probed on a key reconstructed from product names; that probe and its limits are reported in 4.3.

Each probe keeps only classes with enough rows to be estimable, since a class with 20 members contributes noise rather than signal: at least 100 rows for the laboratory probe and 150 for the producer probe, the latter higher because producers are far more numerous. These floors, applied after the 30 conflicting rows are dropped, are why the probes in 3.1 run on 11 of 12 laboratories (37,276 rows) and 33 of 96 producers (30,416 rows) rather than on the full dataset. The probes receive the same 19 value columns as the main model, with empty cells left missing, so a laboratory's analyte panel is visible to them alongside its measured values. The laboratory probe therefore measures instrument and reporting convention together, which is the quantity that matters for a split that shares laboratories; 2.3 measures the reporting convention on its own, and 3.1 reports a missingness-only laboratory probe beside the chemical one.

The probes are multi-class, 11 and 33 classes, and the softmax boosting diverges on them at scikit-learn's default learning rate of 0.1: the laboratory probe's five folds score 40.6 to 69.4 and the producer probe's 10.4 to 68.7. At a learning rate of 0.05 the fold scores span 0.6 points for the laboratory and 1.6 for the producer, and those are the figures reported in 3.1. The binary extraction task is less sensitive: its random split scores 81.0% at the default rate and 79.0% at 0.05, with fold spreads of 0.4 and 0.5 and the held-out score at 54.0% and 54.1%, and it keeps the default.

3. Results

3.1 The alarm fires first

Table 8. Alarm probes: predicting metadata from the 19 analyte columns, random five-fold.

target	classes	rows	accuracy	baseline	lift	bal_acc (chance)
lab	11	37,276	92.3% (+/-0.2)	40.3%	+52.0 pts	76.3% (9.1%)
producer	33	30,416	67.1% (+/-0.6)	14.4%	+52.7 pts	53.4% (3.0%)

The chemical profile identifies the testing laboratory on 92.3% of rows, 52.0 points above the majority baseline, and the producer on 67.1%, 52.7 points above its own. Part of the laboratory signal is the reporting panel: a lookup table on the 13 missingness patterns alone, with no measured value, reaches 57.3% on the same folds, so the measured values add a further 35 points. The fingerprint is not producer memorisation either: grouped by producer on the seven laboratories with at least 100 rows that record one (33,161 rows), the laboratory probe scores 85.4% (+/-3.3) against a 45.3% baseline. In hindsight this is unsurprising, since instruments, calibration, reporting conventions and clientele all differ between laboratories, and this probe cannot separate them. The consequence is that any score obtained while train and test share laboratories is partly reading a laboratory-specific signal. With the split pairs of 2.1 merged the laboratory probe is unchanged at 92.3%, on ten laboratories and a 41.9% baseline.

3.2 Market data: 78.5% becomes 54.0%

Table 9. The reported model under each split scheme.

split scheme	balanced_acc	macro_F1	folds
most-frequent baseline	50.0%	--	--
stratified baseline	50.7%	--	--
random (optimistic)	81.0% (+/-0.4)	81.8%	5
unseen-producer	54.0% (+/-5.6)	50.3%	5
unseen-lab	68.3% (+/-18.6)	67.6%	8
dual-class + unseen	54.6% (+/-2.6)	50.7%	5

The `random` row runs on all 37,344 usable rows and the two producer schemes only on rows that name a producer, so the table's 81.0% and 54.0% are not like with like. On the same 33,229 rows the random split scores 78.5%, and that pair, 78.5% against 54.0%, a gap of **24.4 points** on the unrounded means, is the headline of this paper. The 81.0% includes the 4,115 producer-less rows, which carry the reporting-panel shortcut of 2.3 and the 1,303 samples stored twice (2.1); with those pairs merged it is 80.4%.

How far above chance the held-out score sits depends on how it is measured. The reported model is unweighted and the classes are roughly 1:2, so held out it answers mostly with the majority class (Table 12), and balanced accuracy at the default threshold understates what its scores contain. Table 10 adds ROC AUC and class weighting.

Table 10. Four models on the rows that name a producer; pooled AUC with its producer bootstrap 95% interval.

producer rows (33,229) model	random split		producer held out		pooled AUC [95%]
	bal_acc	AUC	bal_acc	AUC	
boosting, unweighted	78.5%	0.894	54.0% (+/-5.6)	0.625	0.610 [0.535, 0.698]
boosting, class-weighted	80.6%	0.894	55.8% (+/-5.5)	0.621	0.630 [0.551, 0.715]
logistic, unweighted	65.3%	0.753	54.3% (+/-6.2)	0.634	0.604 [0.519, 0.696]
logistic, class-weighted	69.4%	0.752	61.1% (+/-5.1)	0.636	0.662 [0.571, 0.745]

Three things follow. First, a signal does transfer across producers: every model ranks held-out rows with a fold-mean AUC of 0.62 to 0.64, and the producer bootstrap interval of the pooled AUC excludes 0.5 for all four. Second, it is modest. Held out, the reported model keeps less than a third of the above-chance AUC the random split shows, 0.125 of 0.394. Its balanced accuracy of 54.0% is 51.6% when the predictions are pooled, with a producer bootstrap interval of [47.3, 56.6] for the pooled score and [50.2, 58.9] for the fold mean. Third, the size of the loss depends on the estimator and on class weighting: 24.4 points for unweighted boosting, 24.8 with class weights, 11.0 for an unweighted logistic regression and 8.3 for a class-weighted one, whose held-out 61.1% is the highest score any model reaches. The bootstrap resamples the 96 producers with replacement and recomputes the statistic from the held-out predictions of the reported partition. The models are not refitted, so it reflects which producers are tested, not which are trained on.

The reported partition is one draw. Twenty random reassignments of producers to folds (Figure 1b) give the reported model means from 51.3% to 56.6% (median 53.8%, sd 1.7) and the class-weighted one 53.0% to 59.9% (median 55.5%, AUC 0.602 to 0.664); none falls to 50.0%. The five folds of the reported partition score 48.6, 57.6, 46.7, 55.7 and 61.5, a t-interval of [46.3, 61.8] over folds that share training data (p

= 0.22), which is why the producer bootstrap is the better guide. Laboratory identity alone, as a lookup under the same folds, scores 43.6%, below 50 because a laboratory's majority class among the training producers is a poor guide to the held-out ones, so the points above chance are not a laboratory prior.

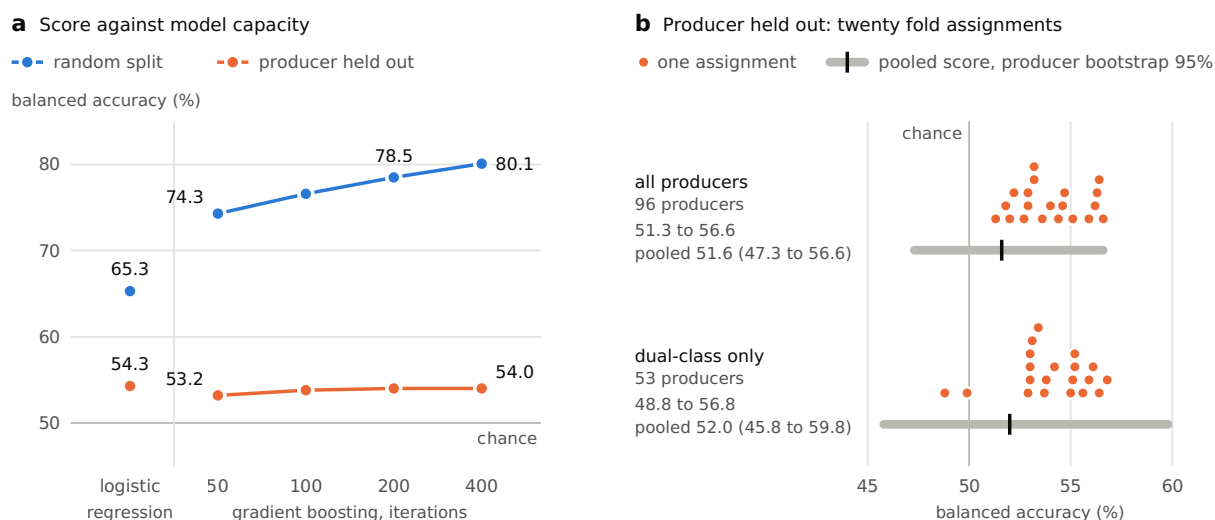


Figure 1. Balanced accuracy on the rows that name a producer, unweighted models. (a) Random split and producer-held-out score, for a standardised logistic regression and for gradient boosting at four iteration counts. (b) Mean score of each of twenty random assignments of producers to folds, for all producers and for the dual-class subset; the bar is the producer bootstrap 95% interval of the pooled score on the reported partition and the tick is that pooled score. Every plotted value is parsed from the evidence logs by `src/evaluation/make_figure.py`.

Table 11. Robustness of the gap. Balanced accuracy; gap in points on unrounded means.

check (producer rows, reported model)	random	held out	gap
reported: 200 boosting iterations	78.5%	54.0%	24.4
random-split seeds 1, 7, 12345			24.6, 24.5, 24.6
50 iterations	74.3%	53.2%	21.1
100 iterations	76.6%	53.8%	22.7
400 iterations	80.1%	54.0%	26.1
1,813 duplicate rows dropped (all rows)	80.8%	55.2%	25.6
rows not named distillate (19,166)	84.1%	60.6%	23.5
rows named for a process (4,345)	77.0%	60.3%	16.8
temporal split (latest 20% as test)	73.4%		
positive control: named distillate or not	89.6%	76.4%	13.2

Table 11 varies what could be suspected of producing the gap. It is stable under reseeding of the random split. It grows with model capacity while the held-out score does not move (Figure 1a), so 24.4 is one point on a curve from 21.1 to 26.1. Duplicate rows do not produce it. A temporal split, training on the earliest 80% of dated rows with 99.3% of its test rows from producers seen in training, scores 73.4%, so time alone costs far less than holding out the producer, although the label and the panel both drift over the period (2.1, 2.2). Two narrower definitions of the label keep more when held out, near 60% on a single partition each with fold spreads of 8.7 and 6.2: the rows not named distillate, and the rows whose names carry a mechanical word in the non-solvent class or a hydrocarbon word in the solvent-based class, the closest this corpus comes to rosin against BHO. The positive control shows that the hold-out does not destroy a score by itself: whether a row is named distillate, a target with a known chemical signature, is recovered at 76.4% for unseen producers.

The strictest test agrees on the size of the loss. The dual-class subset scores 54.6% (+/-2.6) held out against 80.3% (+/-0.7) for a random split on the same 27,751 rows, a gap of 25.6 points on the unrounded means, and its AUC falls from 0.899 to 0.632. On this subset the surviving signal is not separable from chance at the producer level: twenty reassignments give means from 48.8% to 56.8% (median 54.0%), two of them at or below 50.0%, and the producer bootstrap spans [45.8, 59.8] for the pooled balanced accuracy and [0.497, 0.704] for the pooled AUC. On the reported partition its five folds are 54.1, 58.5, 53.2, 56.4 and 50.9, a fold interval of [51.0, 58.3] that excludes 50.0% ($p = 0.024$), which is specific to that partition. Laboratory identity alone scores 40.8% under the same folds. Macro F1 is 50.3% (+/-4.3) and 50.7% (+/-3.1) on the two schemes, against 39.8% and 38.4% for the majority predictor and 49.6% and 49.4% for a stratified shuffle on each scheme's own rows, so on that thresholded metric neither beats a stratified guess.

The confusion matrices show where the apparent skill was concentrated:

Table 12. Pooled confusion matrix of the reported model, producer held out.

unseen-producer	true hydrocarbon: 16,149 correct / 5,818 wrong
	true solventless: 3,332 correct / 7,930 wrong

Solventless recall is **32.3%** as the mean of the five per-fold recalls, the figure the fold-averaged table above is built from. Pooling the matrix instead gives $3,332 / 11,262 = \mathbf{29.6\%}$; the two differ because the producer-grouped folds carry very unequal numbers of solventless rows, from 951 to 4,413, and the fold with the fewest carries the highest recall (48.5% against 25.3% in the fold with the most). We quote both so the matrix and the percentage can be reconciled; the pooled matrix gives a balanced accuracy of 51.6%. For comparison, the per-class recalls are 90.7% and 71.3% under the random split, and 69.9% and 39.3% on the dual-class subset grouped by producer. Either way the model mostly answers with the majority class and its residual skill lives on the easy side. A single balanced-accuracy number would have concealed this; per-class recall does not.

The `unseen-lab` scheme deserves separate comment. Its mean, 68.3%, is the highest of the three group-held-out schemes, but its fold-to-fold standard deviation is **18.6 points**. Performance swings wildly with which laboratory is held out. That is the signature of a model keyed to laboratory convention rather than to a stable chemical difference, and it is consistent with the alarm in 3.1. A study reporting only the mean of this scheme would claim 68.3% for something that is not reproducible across laboratories.

The per-laboratory class balance shows why the spread is so large. Across the eight held-out laboratories the hydrocarbon share ranges from 13.5% (`Cannalytics RI`, 758 rows) to 78.4% (`PureVita`, 3,137 rows), against an overall 65.8%. Holding out a laboratory therefore shifts the test-set prior as well as the instrument, and the two effects cannot be separated with this data. With the split pairs of 2.1 merged the scheme scores 67.4% (+/-19.3). We report it for completeness and do not rest any conclusion on it.

3.3 Controlled experiment: the same direction, an estimator-dependent size

Table 13. Controlled experiment, gradient boosting.

split scheme	balanced_acc	macro_F1	folds
chance level (3 classes)	33.3%	--	--
random 5-fold	56.7% (+/-9.2)	55.7%	5
leave-one-variety-out	38.9% (+/-12.6)	35.9%	6

With gradient boosting, a random split suggests method is somewhat separable at 56.7%, and holding out an entire variety gives **38.9%**, 5.6 points above chance with a fold spread of 12.6 that straddles it. That is the market-data pattern with the confounder changed from producer to variety. It does not survive a change of estimator intact:

Table 14. Controlled experiment, logistic regression.

logistic regression	balanced_acc
random 5-fold	54.3% (+/-2.5)
leave-one-variety-out	46.3% (+/-10.2)

A standardised logistic regression on the same folds drops 8.0 points instead of 17.8, to a score that is above chance: 1 of 1,000 label shuffles reaches 46.3% ($p = 0.002$). Four features and about 130 training samples are a setting where a boosted ensemble can overfit each variety and a linear model cannot, so part of the boosted collapse belongs to the estimator. What both estimators agree on is the direction, and that the method signal which transfers across varieties is modest.

The variety is easier to read than the method:

```
chemistry -> variety: balanced_acc 60.7% (+/-8.3) chance 16.7%
```

The four cannabinoids predict the **variety** at 3.6 times chance with gradient boosting (60.7%) and 2.8 times with logistic regression (46.9%), against 1.2 and 1.4 times chance for the method under leave-one-variety-out. Even in an experiment built to isolate method, with clean HPLC numbers, the variety, which here also means the batch of starting material, is read more easily than the method.

Three checks qualify the boosted figures, each recomputed from the raw file. First, the random-split score depends on the seed: over 200 seeds, five-fold scores average 54.3% (sd 2.7) and the reported 56.7% sits at the 80th percentile, so the seed-robust gap to leave-one-variety-out is 15.4 points rather than 17.8; for logistic regression the 200-seed average is 51.8% (sd 2.3) and the gap 5.5. Second, the drop comes from the variety structure and not from the fold count: holding out six groups of 27 samples assigned at random, which keeps the fold geometry and breaks the variety link, averages 54.0% over 200 draws and never falls to 38.9%. Third, a permutation test over 300 global label shuffles, which also break the balanced design, gives $p = 0.0033$ for the random split, the floor for 300 shuffles, and $p = 0.11$ for leave-one-variety-out, so the boosted grouped score is not distinguishable from chance.

This matters for interpretation. The Cannlytics result alone could be dismissed as a consequence of messy market data. The controlled experiment shows the same direction on clean data, and it also shows that the size of a collapse is a property of the estimator as well as of the data.

4. Discussion

4.1 What this establishes

Two claims, of different kinds.

The **chemical claim** is narrow: the analyte panels of this corpus, 19 columns of which three are derived totals, heavily left-censored, carry only a weak signal of the declared extraction category, non-solvent against solvent-based (2.1), for a producer the model has never seen: an AUC of 0.62 against 0.89 under a random split, and 54 to 61% balanced accuracy depending on the estimator and on class weighting. Producer variation dominates whatever method signal exists: on the dual-class rows under a random split, a classifier given only an encoded producer ID reaches 84.4% and the chemistry 80.3%, while the chemistry grouped by producer reaches 54.6%. Cultivar plausibly does the same, but the two datasets disagree on the evidence and we do not claim it for the market corpus: in the controlled experiment the four cannabinoids read variety at 60.7% against 16.7% chance while reading method at 38.9% against 33.3% (46.9% and 46.3% with logistic regression), whereas in the market data the chemistry reads our reconstructed strain key only weakly, 9.9 points above its baseline (4.3).

The **methodological claim** is general and, we think, the more useful one. Cannabis chemical data has group structure, and the validation protocols in use ignore it. Two independent datasets, one market and one designed, both lose performance when the group is held out: on the market data 24 points of balanced accuracy with gradient boosting and 8 to 11 with logistic regression, and 0.27 of AUC for the reported model; on the designed experiment 15 to 18 points with gradient boosting and 5 to 8 with logistic regression. What the random split rewards is largely group identity, which the same profiles carry strongly: producer at 67.1% against a 14.4% baseline, laboratory at 92.3% against 40.3%, variety at 3.6 times chance.

This gives a concrete magnitude to the concern raised abstractly in [8]: in this corpus the inflation is 24 points of balanced accuracy for a flexible estimator, 27 if rows without a producer are counted on the random side only, and 15 to 18 on a controlled experiment; for a linear model it is 8 to 11 and 5 to 8. It is not a rounding error for either, and it is not the whole score either: the collapse stops short of chance.

4.2 What this does not establish

Not that extraction method leaves no chemical trace. Controlled studies with fixed genetics and fixed instrumentation measure method effects directly, and 3.3 reproduces one such effect under a random split; this work does not contradict them.

Not that a better model would fail. Only that these features do not generalise across producers. A richer assay, a full terpene panel or an LC-MS fingerprint, is a different experiment.

Not that the labels are wrong, and not that they are clean. They are source-declared and agree with the name-derived path on 99.3% of the 11.5% of rows where both exist; that check covers eight rosin and hydrocarbon words only. Product names cannot count mislabelled rows (2.1), and the treatment of ethanol extracts changed within the period (Table 2). What the labels mean is the regulator's category as applied (2.1), which is wider than the rosin-and-hash reading of "solventless": 3.2 gives the result on the rows that come closest to rosin against BHO, and a reader who wants that comparison specifically should rely on those and not on the headline.

Not chance. The held-out scores are low, but a producer-level bootstrap puts the reported model's AUC above 0.5, and a class-weighted linear model reaches 61.1%. The claim is about the size of the inflation, not that nothing transfers.

Not sensitive to the 64 pre-roll rows. Removing them moves random-split balanced accuracy from 81.0% to 80.6% and leaves unseen-producer unchanged at 54.0%, so the 27-point gap becomes 26.6. The conclusion does not depend on their inclusion.

Not a joint hold-out. No scheme here holds out the producer and the laboratory together: the two largest laboratories hold 83% of the rows that carry a producer, so such a split cannot be built from this corpus.

Not a claim about the correctness of the conclusions in [1], [2] or [3]. Their chemical findings may well hold. What we show is that the reported performance figures are obtained under validation that leaves group structure intact, a family of protocols in which we measure substantial inflation for the random-split member, and so cannot be read as estimates of performance on a new producer, genotype or laboratory.

4.3 The confounder we could not test cleanly

Strain is the obvious remaining candidate. Extraction routes are not applied to the same cultivars at the same rates, so the declared category may partly encode which cultivar was used, and cultivar drives terpene profile directly.

We could not test it cleanly. The `strain_name` and `strain_type` columns exist in the source file but are populated for **0 of 37,344** usable rows. The only remaining cultivar signal is inside `product_name`, and extracting a strain key from it leaves a key too dirty to carry a confounder probe:

Table 15. What a strain key extracted from product names contains.

usable rows	37,344	
no cultivar key extractable at all	4,717	(12.6%)
key extracted	32,627	(87.4%)
of those, an obvious non-cultivar word (oil, thc, lab sample, acres, ethanol, raw)	7,386	(22.6% of keys)
rows yielding no usable cultivar signal	12,103	(32.4% of usable)

So roughly a third of the usable corpus carries no cultivar signal, and nearly a quarter of the keys that do parse are packaging or process words rather than cultivar names. A probe built on a key that dirty cannot distinguish "cultivar leaks" from "my regex leaks".

We ran it anyway, and report the output rather than withholding it. Two of the figures are grouped method results: grouping the method task by strain key gives 66.2% balanced accuracy (+/-1.6), and restricting to the 214 keys that appear in both classes, 9,598 rows, gives 56.9% (+/-11.4) with a macro F1 of 42.8%. A third figure is a different quantity, an alarm probe in the sense of 2.6: predicting the key itself from chemistry, on the 109 keys with at least 30 rows (12,829 rows), gives 32.4% against a 22.5% baseline, a lift of 9.9 points. The majority key is the non-cultivar word `oil`, and lifts over 109 and 11 classes are not commensurable, so we read this only as a weak signal.

Both grouped figures fall below a random split on their own rows, 79.2% for the keyed rows and 83.6% for the dual-class keys, and the dual-class one, at 6.9 points above chance, lands near the 4.0 and 4.6 points that the producer-grouped schemes reach. The unrestricted strain grouping is the exception: 66.2%

sits 16.2 points above chance, four times the producer gap, so it does not collapse the way the producer schemes do and we do not present it as though it did. We rest no conclusion on any of the three, because a 22.6%-contaminated grouping key makes the fold composition partly arbitrary, and an 11.4-point fold spread is instability rather than signal. The honest status of the strain question is open and unanswerable with this corpus alone.

4.4 Recommendations

For anyone fitting a classifier to cannabis chemical data:

1. **Group by the entity production will hand you fresh.** Producer, genotype, laboratory. `GroupKFold` and `LeaveOneGroupOut` cost nothing to run.
2. **Report the random-split score alongside the grouped one.** The gap is the informative quantity and is cheap to obtain.
3. **Never report plain accuracy without its majority-class baseline.** Reference [1]'s 95.2% sits on a 78.8% baseline, so the lift is 16.4 points.
4. **Report per-class recall and the between-fold spread.** Our `unseen-lab` mean of 68.3% carries a standard deviation of 18.6; the mean alone is misleading.
5. **Run an alarm model before the real task.** If the chemistry predicts the laboratory, any laboratory-sharing score is partly laboratory-specific.
6. **Keep a tested-versus-detected distinction in censored panels.** Merging a measured zero with a missing cell changes the coverage gap in our data from 1 to 6 points to 18 to 25.
7. **Report the gap for more than one estimator.** Here it runs from 21 to 26 points across boosting iterations and is 8 to 11 for a linear model.
8. **Report a threshold-free metric and a class-weighted model.** Our unweighted model's held-out 54.0% hides an AUC of 0.62; class weighting alone moves a linear model from 54.3% to 61.1%.

4.5 Where a decisive answer would come from

The remaining untested avenue is a method-comparison dataset pairing extraction method with a full metabolomic fingerprint including terpenes, such as the MassIVE LC-MS/GC-MS collections. That alone could distinguish "method leaves no usable trace" from "method leaves a trace, but not in the 4-cannabinoid or 19-analyte projections tested here". It is a mass-spectrometry corpus of several gigabytes and a separate build, not an extension of this work.

5. Conclusion

The declared extraction category, non-solvent against solvent-based, is only weakly identifiable from the cannabinoid and terpene panels of this corpus of public certificates of analysis for an unseen producer. On the rows that name a producer, a random split reports 78.5% balanced accuracy and an AUC of 0.89; holding out the producer leaves 54.0% (51 to 57% across twenty fold assignments) and an AUC of 0.62, whose producer-level bootstrap interval, 0.54 to 0.70, excludes chance. Less than a third of the above-chance skill survives. A designed HPLC experiment shows the same direction when variety is held out,

56.7% to 38.9% with gradient boosting and 54.3% to 46.3% with logistic regression, against 33.3% chance. The same profiles identify the producer at 67.1% against a 14.4% baseline, the laboratory at 92.3% against 40.3%, and the variety at 3.6 times chance.

The published accuracies cited here, 95% and 100%, are obtained under validation that holds out no group, the family of protocols for which we measure 24 points of inflation on market data and 15 to 18 on a designed experiment with gradient boosting, and 8 to 11 and 5 to 8 with a linear model. We ran random cross-validation, not Kennard-Stone selection, so the magnitude we report is specific to the former; what the two share is that neither holds out the producer, the genotype or the laboratory. We do not claim the published chemical conclusions are wrong, and we did not rerun those studies. We claim their numbers do not answer the question a reader will assume they answer. Group-aware validation is not a refinement here. It changes the size of the result severalfold.

Data and code availability

Source data are public and are not redistributed here. The Cannlytics `cannabis_results` corpus [9] and Zenodo record 13823859 [10] are both CC-BY-4.0 and are retrieved by the ingestion scripts described below, which verify the downloaded size against the figure declared by the host API.

All analysis code and the raw evidence log for every run described in this manuscript are public at <https://github.com/ArtBreguez/cannabis-extraction-ds> (code MIT, manuscript CC-BY-4.0). The structure is:

<code>src/ingestion/</code>	retrieval with download-integrity verification
<code>src/preprocessing/</code>	labelling, unit checks, non-detect encoding
<code>src/models/</code>	the four split schemes and the alarm probes
<code>src/evaluation/</code>	recomputation and audit of every figure reported here
<code>docs/evidence/</code>	raw stdout of each run, committed unedited

A single script, `src/evaluation/rederive_paper_numbers.py`, recomputes every dataset figure in this manuscript directly from the labelled dataset, and `src/evaluation/audit_preprint.py` checks each claimed number in this text against either that recomputation or the committed log, failing on any disagreement.

Author contributions

A.G.B. is the sole author and is responsible for the conceptualisation, data curation, formal analysis, methodology, software, validation, visualisation and writing of this work. The assistance of AI tools is described below.

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Competing interests

The author produces cannabis-related content on social media under the handle @HiddenTerps and has a commercial interest in mechanically separated (solventless) extracts, which fall inside the non-solvent category studied here. This work reports that the category is only weakly recoverable from routine laboratory panels, which runs against that interest rather than supporting it.

Use of AI tools

AI coding assistants were used throughout this work, under the author's direction and review. Their contribution was to the analysis code, the verification harness and the manuscript prose: writing and refactoring the scripts in `src/`, running the audit passes described above, and drafting and revising text that the author then checked against the evidence logs. Several defects corrected before posting were found by that review loop, among them a leakage pre-check that measured six analyte columns while the model saw nineteen, a percentage whose numerator and denominator described different populations, and two multi-class alarm probes whose reported lifts were artefacts of a diverging learning rate (2.6).

AI tools do not meet authorship criteria and are not listed as authors. The author is responsible for every claim, number and conclusion in this manuscript. Every dataset figure is recomputed from the labelled data by `src/evaluation/rederive_paper_numbers.py`, every model figure is read from the committed log of the run that produced it, and `src/evaluation/audit_preprint.py` asserts each against this text, exiting non-zero on any disagreement, so no published value rests on an assistant's unverified output.

A note on provenance

An early version of this dataset audit concluded the research was not viable, reporting 6,336 rows and 24 versus 11 labellable samples. That measurement was made on a download truncated to 7% of its size, 177.9 MB of 2,534.8 MB, cut mid-record. `curl` reported success, the parser opened the file and discarded a single malformed line, and every downstream number was plausible. The truncation was found by comparing the local file size against the figure declared by the host API. Every number in this manuscript comes from the verified intact file, and download integrity is checked against the source in `src/ingestion/`.

References

- [1] Freeman, T. P., Beeching, E., Craft, S., Di Forti, M., Frison, G., Lindholm, C., Oomen, P. E., Potter, D., Rigter, S., Rømer Thomsen, K., Zamengo, L., Cunningham, A., Groshkova, T., Sedefov, R. Applying machine learning to international drug monitoring: classifying cannabis resin collected in Europe using cannabinoid concentrations. *European Archives of Psychiatry and Clinical Neuroscience*, 275(2), 421-429, 2025. doi:10.1007/s00406-024-01816-w. PMC11910419.
- [2] Tran, J., Vassiliadis, S., Elkins, A. C., Cogan, N. O. I., Rochfort, S. J. Rapid In Situ Near-Infrared Assessment of Tetrahydrocannabinolic Acid in Cannabis Inflorescences before Harvest Using Machine Learning. *Sensors*, 24(16), 5081, 2024. doi:10.3390/s24165081.

- [3] Birenboim, M., Chalupowicz, D., Maurer, D., Barel, S., Chen, Y., Fallik, E., Paz-Kagan, T., Rapaport, T., Sadeh, A., Kengisbuch, D., Shimshoni, J. A. Multivariate classification of cannabis chemovars based on their terpene and cannabinoid profiles. *Phytochemistry*, 200, 113215, 2022. doi:10.1016/j.phytochem.2022.113215.
- [4] Kaufman, S., Rosset, S., Perlich, C., Stitelman, O. Leakage in data mining: formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4), 15, 2012. doi:10.1145/2382577.2382579.
- [5] Kapoor, S., Narayanan, A. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9), 100804, 2023. doi:10.1016/j.patter.2023.100804.
- [6] Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Aroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., Dormann, C. F. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8), 913-929, 2017. doi:10.1111/ecog.02881.
- [7] Guignard, F., Ginsbourger, D., Levy Häner, L., Herrera, J. M. Some combinatorics of data leakage induced by clusters. *Stochastic Environmental Research and Risk Assessment*, 38(7), 2815-2828, 2024. doi:10.1007/s00477-024-02715-1.
- [8] Király, P., Tóth, G. Being Aware of Data Leakage and Cross-Validation Scaling in Chemometric Model Validation. *Journal of Chemometrics*, 39(4), e70026, 2025. doi:10.1002/cem.70026.
- [9] Cannlytics. `cannabis_results`: public cannabis lab test results. Hugging Face Datasets, https://huggingface.co/datasets/cannlytics/cannabis_results, revision 8724df97f3c40b3f844157606dd91e97509a8cb9 (last modified 2026-02-02), file `all-results-latest.csv`, retrieved 2026-09-21. CC-BY-4.0.
- [10] Solís García, H. F., Suntaxi Crisanto, S. L. F., Vargas Delgado, L. F., De la Rosa Martínez, A. F., Londoño Larrea, P., González Benítez, D., Montúfar Delgado, C. Yields, Cannabinoids Quantification, and Predictive Programming Codes Using Machine Learning for Non-Psychoactive Cannabis Flowers and Extracts (*Cannabis sativa* L.) Cultivated in Ecuador. Zenodo, 2024. CC-BY-4.0. doi:10.5281/zenodo.13823859.
- [11] Ambroise, C., McLachlan, G. J. Selection bias in gene extraction on the basis of microarray gene-expression data. *Proceedings of the National Academy of Sciences*, 99(10), 6562-6566, 2002. doi:10.1073/pnas.102102699.
- [12] Nevada Cannabis Compliance Board. Nevada Cannabis Compliance Regulations, Regulation 11, section 11.050, Required quality assurance tests; submission of wet cannabis for testing. Quoted from the version of 21 July 2022, https://ccb.nv.gov/wp-content/uploads/2022/07/Reg-11_v072122.pdf; the same wording stands in the version of 24 October 2023, https://ccb.nv.gov/wp-content/uploads/2023/11/Reg-11_v102423.pdf. The regulations effective November 2020, <https://ccb.nv.gov/wp-content/uploads/2021/02/Effective-NCCR-as-of-Nov-2020.pdf>, carry the same solvent-based definition and list terpene analysis for both categories. All accessed 2026-10-06. The 2018 predecessor, NAC 453D.780, named CO2 only.

[13] Kennard, R. W., Stone, L. A. Computer Aided Design of Experiments. *Technometrics*, 11(1), 137-148, 1969. doi:10.1080/00401706.1969.10490666.